SIC - MMAB: Synchronisation Involves Communication in Multiplayer Multi-Armed Bandits

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Summary

- Synchronisation $\implies$ communication through collisions in MMAB.
- Contradicts previous lower bounds.
- Synchronisation assumption is unrealistic and a loophole in the model.
- More realistic dynamic model with limited feedback: first logarithmic regret algorithm.

Multiplayer bandits model

Motivation cognitive radio networks

Framework K arms, $M \leq K$ players. At each $t \leq T$, player j pulls $\pi^j(t)$

Reward $r^j(t) := X_{\pi^j(t)}(t)(1-\eta_{\pi^j(t)}(t))$

- collision indicator $\eta_k(t) := \begin{cases} 1 \text{ if several players pull } k, \\ 0 \text{ otherwise} \end{cases}$
- $X_k(t) \in [0,1]$ *i.i.d.* with $\mathbb{E}[X_k] = \mu_k$

Decentralized no information exchange between players

Observations **Collision sensing:** observes $r^j(t)$ and $\eta_{\pi^j(t)}(t)$
No sensing: observes $r^j(t)$

Synchro models **Static:** any player plays for $t = 1, \dots, T$
Dynamic: player j plays for $t = 1 + \tau_j, \dots, T$

Goal with $\mathbf{M}(t)$ set of players at time t, minimize regret

$$R_T := \sum_{t=1}^{T} \sum_{k=1}^{\#\mathbf{M}(t)} \mu_{(k)} - \mathbb{E}_\mu \left[\sum_{t=1}^{T} \sum_{j \in \mathbf{M}(t)} r^j(t) \right]$$

Notation $\mu_{(1)} \geq \dots \geq \mu_{(K)}$ and $\bar{\Delta}_{(M)} := \min_{i=1,\dots,M} (\mu_{(i)} - \mu_{(i+1)})$

State of the art bounds

Setting	Prior knowledge	Asympt. Upper bound
Centralized Multiplayer	M	$\sum_{k>M} \frac{\log(T)}{\mu_{(M)}-\mu_{(k)}}$ [3]
Decentralized, Col. Sensing	$\mu_{(M)}-\mu_{(M+1)}$	$\frac{MK\log(T)}{(\mu_{(M)}-\mu_{(M+1)})^2}$ [4]
Decentralized, Col. Sensing	-	$\sum_{k>M} \frac{\log(T)}{\mu_{(M)}-\mu_{(k)}} + MK\log(T)$
Dec., Col. Sensing , Dynamic	$\bar{\Delta}_{(M)}$	$\frac{M\sqrt{K\log(T)T}}{\bar{\Delta}^2_{(M)}}$ [4]
Dec., No Sensing, Dynamic	-	$\frac{MK\log(T)}{\bar{\Delta}^2_{(M)}} + \frac{M^2K\log(T)}{\mu_{(M)}}$

Our results in red.

SIC-MMAB: static with collision sensing

Observation: collision indicator $\eta_k(t) \in \{0,1\} \to$ a bit sent from one player to another.
$\implies$ communication possible between players.

Algorithm 1: SIC-MMAB
Initialization Phase: estimate M and player rank j in time $\mathcal{O}(K\log(T))$
for $p = 1, \dots, \infty$ **do**
  Exploration phase: explore each active arm 2^p times without collision
  Communication phase: players exchange statistics of arms using collisions
  Eliminate suboptimal arms
  Attribute optimal arms to players (who enter exploitation phase)
end
Exploitation phase: pull attributed arm until T

Communication protocol

When player i sends stats of arm k to player j:

- Pull k_j to send a 1 bit to j (collision)
- Pull k_i to send a 0 bit (no collision)
- Send quantized empirical mean in binary

For 2^p exploration rounds, communicate in $p \to$ sublogarithmic comm. regret

Regret of SIC-MMAB

$$\mathbb{E}[R_T] \leq \mathcal{O}\left(\overbrace{\sum_{k>M} \frac{\log(T)}{\mu_{(M)}-\mu_{(k)}}}^{\text{Exploration phases}} + \underbrace{MK\log(T)}_{\text{Initialization phase}} + \overbrace{M^3K\log^2\left[\frac{\log(T)}{(\mu_{(M)}-\mu_{(M+1)})^2}\right]}^{\text{Communication phases}} \right)$$

Contradiction with the lower bound

Centralized lower bound

$$\Omega\left(\sum_{k>M} \frac{\log(T)}{\mu_{(M)}-\mu_{(k)}} \right) [1]$$

SIC-MMAB

Decentralized lower bound

$$\Omega\left(M \sum_{k>M} \frac{\log(T)}{\mu_{(M)}-\mu_{(k)}} \right) ?? \; [2]$$

NO !

Decentralized $\sim$ Centralized

Toward a realistic model?

Unrealistic communication protocols $\to$ loophole in current model.
Which assumption did go wrong?

- Collision sensing? **No:** still send a bit in $\frac{\log(T)}{\mu_{(K)}}$ rounds without it
- Synchronisation? **Yes!** Communication possible because players know when to *talk* to each other

Synchronisation: unrealistic assumption leading to *hacking* algorithms.
Let's remove it $\to$ dynamic model.

DYN-MMAB: dynamic no sensing

Algorithm 2: DYN-MMAB
Exploration phase: pull $k \sim U(K)$
Update **occupied** arms and queue of arms to **exploit**
if $r_k > 0$ *and* $k =$ *arm to exploit* **then**
  Enter exploitation phase
end
Exploitation phase: pull exploited arm until T

How does it work?
Do not estimate μ_k but $\gamma\mu_k$ where $\gamma \simeq \mathbb{P}(\text{collision})$
Still distinguish optimal from suboptimal arms
What if another player exploits k?

- either $\hat{r}_k \to 0$ quickly and arm k becomes suboptimal
- or too many 0s in a row $\implies$ k is occupied

Regret of DYN-MMAB

$$\mathbb{E}[R_T] \leq \mathcal{O}\left(\frac{M^2K\log(T)}{\mu_{(M)}} + \frac{MK\log(T)}{\bar{\Delta}^2_{(M)}} \right)$$

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